Journal of Cleaner Production 455 (2024) 142323

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Assessment of CO_2 abatement potential of heat pumps using pinch analysis for the Swiss chocolate industry

Navdeep Bhadbhade [a,*], Benjamin H.Y. Ong [b], Donald G. Olsen [a], Beat Wellig [a], Martin K. Patel [a]

[a] Energy Efficiency, Department F.-A. Forel for Environment and Aquatic Sciences, Institute for Environmental Sciences (ISE), University of Geneva, Switzerland
[b] Lucerne University of Applied Sciences and Arts, Competence Center Thermal Energy Systems and Process Engineering, Technikumstrasse 21, 6048, Horw, Switzerland

ARTICLE INFO

Handling Editor: Cecilia Maria Villas Bôas de Almeida

Keywords:
Decarbonization
Energy efficiency
Process integration
Heat pump
Energy demand profiles

ABSTRACT

Decarbonizing industrial processes is crucial for achieving net-zero targets, with process heat electrification emerging as a key pathway. Heat pumps (HPs) are pivotal for this transition. This study introduces a novel methodology to construct exemplar profiles for specific energy demands, representative profiles for various production capacities, and energy demand profiles at product, subsector, or sector levels, using Swiss chocolate production as a case study. Exemplar profiles offer flexibility in generating representative profiles based on production capacities, aiding in identifying integration opportunities and sizes for HPs. Sector-wide profiles provide a broader perspective, aiding in visualizing bottom-up sector-wide potential for HPs. Based on the analysis of four representative profiles for Swiss chocolate production, two potential standard heat pump (SHP) and three high-temperature heat pump (HTHP) integration opportunities were identified. Techno-economic data from 30 commercially available HPs were collected, assessing their cost-effectiveness for Swiss chocolate manufacturing using the levelized cost of saved energy and CO_2 abatement. Factors such as company size and electricity and natural gas prices impact the cost-effectiveness of HP integration. Despite variations, HPs with real coefficients of performance above 2.6 generally prove cost-effective for average energy prices and current CO_2 levies in Swiss chocolate production. Bottom-up estimations of HP integration potentials using product-specific energy demand profiles suggest a potential reduction of CO_2 emissions by up to 56% and an improvement in energy efficiency by up to 28% in Swiss chocolate production.

Nomenclature

ANF	Annuity factor
B	Annual benefits of the measure (CHF)
CA	Annual potential CO_2 savings by technology or measure *y* (tCO_2/y)
CCs	Composite curves
CF	Annual cash flow for the year *t* (CHF)
CHF	Swiss Franc
CIP	Cleaning in place
COP	Coefficient of performance
DR	Discount rate
EF	Emission factor of fuel (tCO_2/GJ)
ES	Annual final energy savings achieved by measure (GJ/yr)
GCC	Grand Composite Curve
HP	Heat pump
HTHP	High-temperature heat pump
HVAC	Heating, Ventilation and Air Conditioning
I	Initial investment (CHF)
I_a	Annualized initial investment cost (CHF)
ISSP	Indirect source-sink profiles
IR	Infrared
L	Lifetime of measure *y* (years)
LCSE	Levelized costs of saved energy by technology or measure *y* (CHF cents/kWh)
LCCO2	Levelized costs of CO_2 abatement (CHF cents/kg-CO_2)
NPV	Net present value of measure *y* (CHF)
O&M	Annual operations and maintenance costs of measure *y* (CHF)
PA	Pinch Analysis
P_e	Price of electricity (CHF/GJ) **P_{fi}:** Price of fuel i (CHF/GJ)
P_{CO2}	CO_2 price in year *t* (CHF/t CO_2)
PPP	Purchasing power parity
SEC	Specific energy consumption
SF	Scaling factor
SHP	Standard heat pump

(*continued on next column*)

(*continued*)

(*continued on next page*)

* Corresponding author.
E-mail address: navdeep.bhadbhade@hslu.ch (N. Bhadbhade).

https://doi.org/10.1016/j.jclepro.2024.142323
Received 11 December 2023; Received in revised form 15 April 2024; Accepted 22 April 2024
Available online 23 April 2024
0959-6526/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

(*continued*)

t	year
TES	Thermal energy storage
TFE	Total final energy

1. Introduction

1.1. Background

Industrial process heating accounts for more than 50% of global heat demand (IEA, 2019), making decarbonizing process heat generation a crucial step towards achieving the net-zero targets set by many countries for the middle of this century. Industrial decarbonization can be achieved by improving fuel consumption efficiency, replacing fossil fuels with low-carbon fuels or renewables, electrifying industrial processes, and implementing negative emission technologies. While improvement in fuel consumption efficiency and negative emissions technologies are more suitable for energy-intensive industry sectors with relatively high-temperature process heat requirements (such as iron and steel, cement, and chemicals), process heat electrification is more suited for non-energy-intensive sectors with relatively low process temperatures (such as food and beverage).

In line with many other countries, Switzerland has set a net-zero target for 2050 (BFE, 2023b). Industrial processes responsible for 20% (BFE, 2023a) of total fuel demand and 15% of CO_2 emissions (EEA, 2023) will play an important role in the Swiss energy system's transition towards net-zero emissions. Due to the very low CO_2 emission intensity of electricity generated in Switzerland (i.e., 0.12 kg CO_2/kWh (Krebs and Frischknecht, 2021)) compared to other European countries, electrification of the industrial processes can be a very effective way of industrial decarbonization. Process heat electrification can be achieved by replacing fossil fuels either by direct conversion of electricity to heat (e.g., electric arc furnaces in the steel industry or infrared dryers in the food industry) or by excess heat utilization primarily through the integration of technologies like heat pumps (HP) (IEA, 2022). Utilization of excess heat from industrial processes is technically one of the most impactful ways of improving the efficiency of industrial processes and reducing their CO_2 emissions (Bhadbhade and Patel, 2024) HPs are potentially very convenient and cost-effective electrification options that can be implemented without any significant modifications to the core processes (Jibran et al., 2022).

Since their inception a few decades ago, the Pinch Analysis (PA) tools have been extended to analyze the integration of energy conversion technologies in industrial processes. Townsend and Linhoff (1983) presented heuristics of heat pump integration using process integration principles. Wallin and Berntsson (1994) presented methods to identify the optimum HP integration using the composite curves (CCs) and further extended the methodology to the use of grand composite curves (GCCs) for identifying HP integration opportunities. Klemes et al. (1997) introduced a novel methodology to analyze the energy and CO_2 reduction potentials for total site. HP integration methodologies were further developed to identify optimum integration opportunities based on PA principles and presented using the case studies for various industrial processes. Becker et al. (2012) presented a practical case study of the application of PA and HP integration in a Swiss dairy. Arpagaus (2019) presented a case study of HP integration in the Swiss chocolate factory. Wallerand et al. (2018) developed an optimization framework for HP integration along with the integration of renewables and presented with the case study of Swiss cheese factory. Kapustenko et al. (2008) presented an application of PA for integrating a HP in ammonia refrigeration system at a meat processing facility. Few different methodologies were developed for optimum integration of HPs and thermal energy storage (TES) in non-continuous processes. Olsen et al. (2017) developed a methodology to identify indirect source sink profiles (ISSP). Agner et al. (2021) built on the ISSP methodology and presented a graphical method for the combined integration of HP and TES. Stampfli et al. (2019) built on the previous approach and developed a practical method for HP integration in a non-continuous process. Seevers et al. (2018) developed a methodology for heat pump sizing for fluctuating heat demand. Schlosser et al. (2019) presented a review of sector and process-specific integration concepts for HPs based on the evaluation of GCCs.

Given their important role in industrial decarbonization, there has been a growing interest in the techno-economic analysis of commercially available HPs for various industry sectors. Arpagaus et al. (2018) presented a comprehensive review of commercially available industrial HPs in Switzerland. Schlosser et al. (2020), presented the application, performance and economic feasibility of large-scale industrial HPs for Germany. Jesper et al. (2021) developed methodology for economic and ecological evaluation of the HP and modelled a correlation between the coefficients of performance (COPs) and operating conditions based on the data from commercially available large-scale HPs.

Along with the advancement in the methodologies for HP integration in the process industries, there have been a several notable studies that present the methodologies for the estimation of sector-wide HP integration potentials. Due to the relatively high share of heat demand at rather low temperatures (Zuberi et al., 2018), the food industry subsector is one of the most suitable industry subsectors for HP integration, which has been the focus for several studies. For example, Zuberi et al. (2018) analyzed HP integration potential in the U.S. food industry and concluded that HP integration represents about 20% of the total final energy (TFE) reduction potential. A similar study on the integration potential of high temperature heat pumps (HTHP) in the French food industry was published by Seck et al. (2013), which led to a conclusion that HP integration can potentially reduce up to 13.6 % of TFE demand from the French food industry. In order to estimate the sector-wide theoretical and practical heat integration potentials, Kantor et al. (2018) developed a methodology to construct sector-wide bottom-up GCCs using publicly available generic information about process requirements. Ong et al. (2023) developed an approach to construct a bottom-up sector wide net heat sink and source profiles based on the previously conducted Pinch Analysis of individual companies.

The HP integration potentials estimated in the existing literature are either based on one industrial site or top-down estimates using the generic temperature levels and energy demand statistics. However, the optimum integration and cost-effectiveness of HPs and other energy conversion technologies depend on the accurate quantification of heating and cooling demands for various production capacities and operating cases. There is currently a lack of methodology to efficiently scale the process requirements for different production capacities and operating cases and thus facilitating the precise estimation of potentials for energy conversion technologies and excess heat. In this work, a methodology to develop exemplar profiles (process requirements for one tonne of product), representative profiles (scaling the process requirements for the given operating conditions) and product-specific energy demand profile (process requirements for entire Swiss chocolate production) is presented using Swiss chocolate production as a case study. Thus, the methodology presented in this work facilitates identification of HP integration opportunities for various production capacities and operating cases as well as the bottom-up estimation of HP integration potentials for a broader scope.

1.2. Aims and objectives

Despite being one of the promising electrification technologies and a steady increase in the range of available HP models in the market in recent years, the wider spread of industrial HPs in the industry has been facing barriers such as a lack of awareness about the possible economic opportunities for HP integration, lack of knowledge about commercially available options among potential users (Arpagaus et al., 2018), high

initial capital costs of HP solutions and low expected payback periods by industry decision-makers (Wolf et al., 2017). The tailor-made designs for the HP systems often have higher initial investment costs than the commercially available, off-the-shelf HP options leading to higher payback times (Schlosser et al., 2020). As a result, despite several energy conversion technologies (e.g., solar thermal, combined heat and power (CHP)) are technically feasible for integration in the Swiss chocolate production the current work focuses on identifying HP integration opportunities using the techno-economic data for commercially available HPs.

Given this background the aim of the present work is twofold.

1. To develop a methodology for the exemplar and representative profiles and demonstrate their use to identify the optimum placements and capacities of HPs in the Swiss chocolate production.
2. Develop product-specific energy demand profiles and demonstrate its use to estimate bottom-up HP integration potentials for entire Swiss chocolate production and assess their cost-effectiveness.

The methodology in this work is presented with an objective to equip industrial decision-makers with a tool to efficiently estimate process requirements for the given production capacity and identify the cost-effective off-the-shelf HP options. The methodology presented in this work can also be used to identify the integration potentials for other energy conversion technologies and available excess heat for other non-industrial uses (e.g., district heating networks) as well as for different geographic scopes.

1.3. Structure of chocolate manufacturing industry in Switzerland and process description

In 2021, the overall production in the Swiss chocolate industry reached about 200,000 tonnes. Accounting for about 6% TFE demand of the Swiss food and beverage industry (Bhadbhade and Patel, 2020), the Swiss chocolate manufacturing employs about 4,000 workers and generates about CHF 1.7 billion in annual sales (Chocosuisse, 2021).

1.3.1. Industry structure

Shares of energy carriers in the TFE consumption of chocolate manufacturing in Switzerland are estimated based on the information provided in EnAW database[1] and presented in Fig. 1. The majority of processes involved in chocolate production utilize electricity. As a result, approximately 59% of the TFE demand in the Swiss chocolate industry is covered by electricity. Roasting and debacterization are large heat consumers at high temperatures. Roasting technologies employed in Swiss chocolate manufacturing typically utilize steam or direct natural gas. Thus, the majority of heat generated for use in chocolate manufacturing is produced using natural gas (31% of TFE). District heating (8%), fuel oil (1%), and biogas (<1%) are responsible for relatively small shares of TFE demand.

In order to take the differences in electricity prices paid by Swiss chocolate manufacturers into account, companies in Swiss chocolate manufacturing are categorized into small, medium and large enterprises based on the definitions provided by Elcom (ElCom, 2023). Fig. 2 presents the estimated shares of annual production by companies of various sizes and corresponding Elcom categories.

1.3.2. Process description

A schematic of the generic process for chocolate manufacturing in Switzerland is presented in Fig. 3. The process begins with the cleaning and debacterization of raw materials such as cocoa beans. Depending on

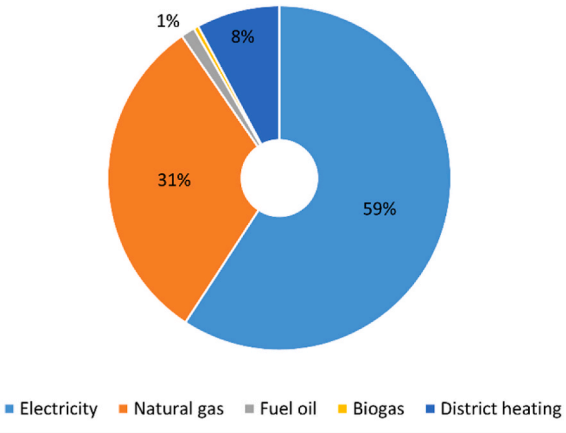


Fig. 1. Energy mix for Swiss chocolate production ((Energie-Agentur der Wirtschaft EnAW, 2016).

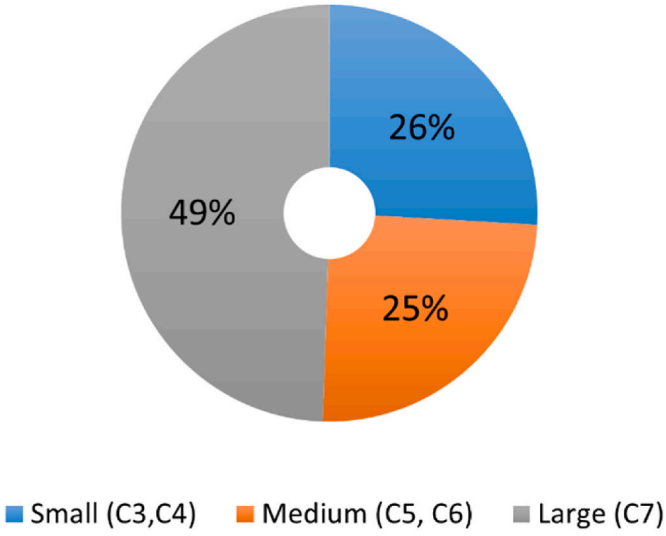


Fig. 2. Share of chocolate production by company size [Company category (ElCom, 2023),company size(FSO, 2023)].

the intended product, cocoa beans are either fed to the breaking and winnowing process (cocoa powder) or directly to the roaster (cocoa liquor). In order to bring out the aroma of the cocoa and remove the moisture, cocoa beans or cocoa nibs (cocoa beans after removal of shells) are roasted at 100 °C–120 °C. Depending on the intended product, roasting occurs either using the hot air produced by the heat exchange with steam (for whole bean roasting) or in a natural gas-heated roaster (for nibs roasting e.g., tornado roaster). Whole roasted beans are lightly crushed to remove the shell from the nibs using toothed rollers or impact rollers. The shells and nibs are then separated using the winnowing process. The nibs are fed to rollers, where they are ground to a fine cocoa paste. The heat from grinding melts the cocoa butter, which is removed using large hydraulic presses. The remainder is cocoa cake which is ground and sifted to produce cocoa powder. The cocoa powder is mixed with cocoa butter, sugar as well as either condensed milk or milk powder in a kneader to prepare a homogenous, paste-like mixture or fed directly to rolling mills. This paste is then further refined to the particle size of 15–20 μm in rolling mills. The desired aroma, texture and flavor are imparted to chocolate through surface scraping and agitation in the process step called conching. The temperature of chocolate paste rises to about 80 °C during the conching process and is then typically maintained at about 55 °C using cooling water. Chocolate mass formed in the conching process is transferred to the tempering and molding process

[1] EnAW stands for "Energie-Agentur der Wirtschaft". The agency was set up by the private sector to improve the energy efficiency of companies and to reduce their CO_2 emissions in the context of a target agreement.

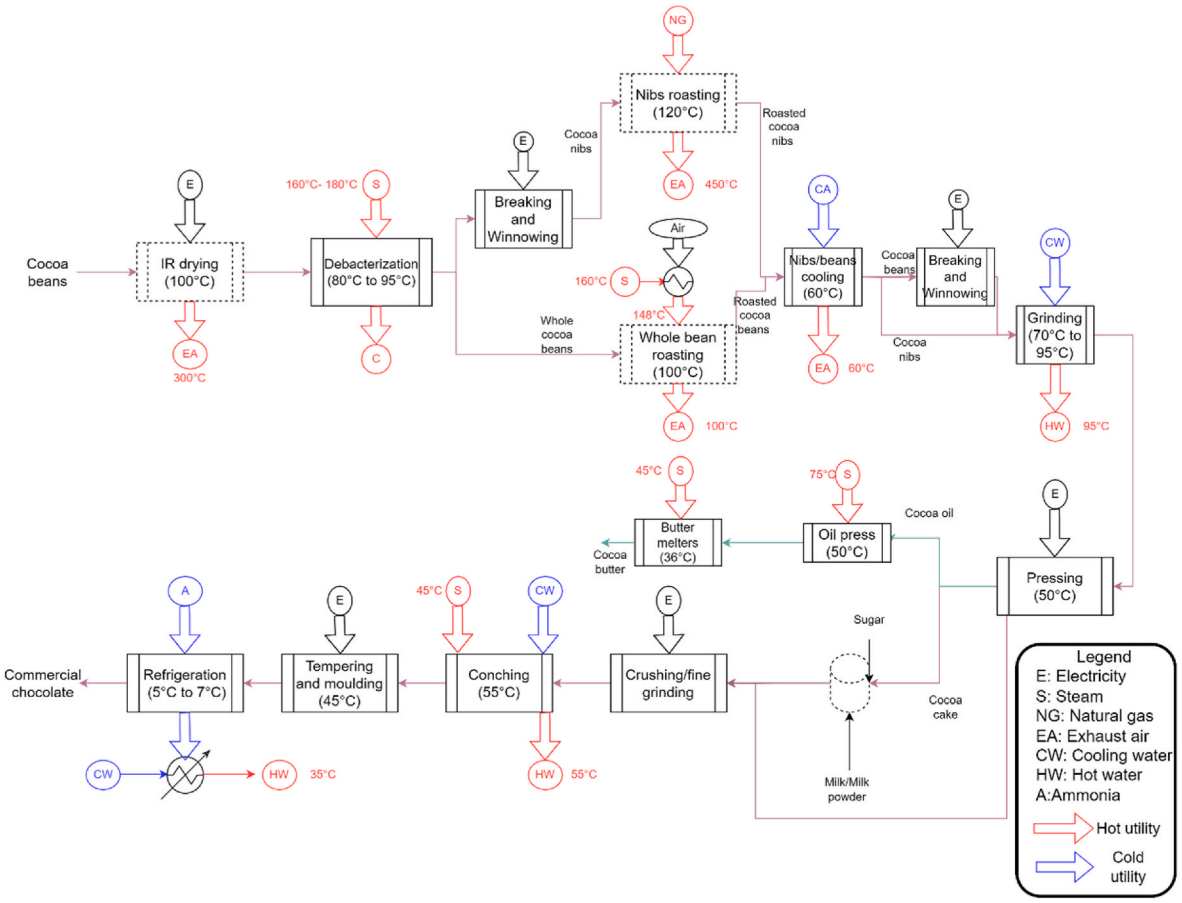


Fig. 3. Schematic of chocolate production process in Switzerland.

which takes place at about 45 °C. Molded chocolate is cooled down to between 5 °C and 7 °C using a refrigeration system. It should be noted that chocolate facilities producing specialty products have dedicated production lines that exhibit some variation in process steps and process conditions described above and depicted in Fig. 3. However, due to data confidentiality and heterogeneity in process conditions, they are not included in the analysis. The product-specific energy demand profile profiles presented below, and the subsequent HP integration analysis are based on the description and generic schematic presented in this section.

Based on the data gathered through various Pinch studies of Swiss chocolate manufacturing companies, approximately 63% of the total heat in chocolate production is required below 55 °C (Meier, 2006). The process steps of conching, blending, tempering and molding require heat below 55 °C. Cleaning in place (CIP) and heating, ventilation and air conditioning (HVAC) represent approx. 15% of the total heat demand at temperatures between 55 °C and 100 °C. Processes of roasting, beans/cocoa paste drying and debacterization require heat above 100 °C and together represent up to about 23% of total heat demand (U. Flükiger and W, 2011)(see Fig. 4).[2]

1.4. Heat pumps

HPs are reverse heat engines that are used to transfer the available heat from lower temperatures (T_{source}) to higher temperature (T_{sink}). HPs typically consist of an evaporator to absorb heat, a compressor to raise the pressure and temperature of the working fluid and a condenser to reject the heat. External energy or work needed by the HP to transfer heat from a lower temperature to a higher temperature is typically provided by electricity (more specifically for closed-system compression HPs).

Table 1 presents the typical classification of HPs based on the maximum T_{sink} that can be achieved by the HP. Moreover, HPs can also be classified as open system and closed system. Open system HPs are further classified as thermal vapor recompression and mechanical vapor recompression based on the technology. Closed system HPs are further classified as sorption and compression HPs based on the technology. Both open circuit and closed loop HPs are implemented in the industry. However, open circuit HPs are confined to niche applications, and they often compete with direct heat integration opportunities using heat exchangers. Due to their flexibility and high supply temperatures, closed loop HPs on the other hand, are most suitable options for industrial applications (Wolf et al., 2017). Thus, current work focuses on the analysis of single stage, closed loop HPs, which are commercially available and for which the implementation data is publicly available.

2. Data and methods

2.1. Chocolate process data

Process stream data for chocolate production such as temperature levels, specific heat and mass flow rates are obtained from publicly available Pinch Analysis studies for chocolate (Meier, 2008) and cocoa (Arnold and Brühlmeier, 2010) manufacturers in Switzerland. Average values for specific final energy consumption (SEC; kWh/tonne) for chocolate manufacturing process steps in Switzerland are estimated based on typical temperature levels, mass flow rates, and annual

[2] It should be noted that the temperature levels presented in Fig. 4 and the accompanying text are based on the replacement streams.

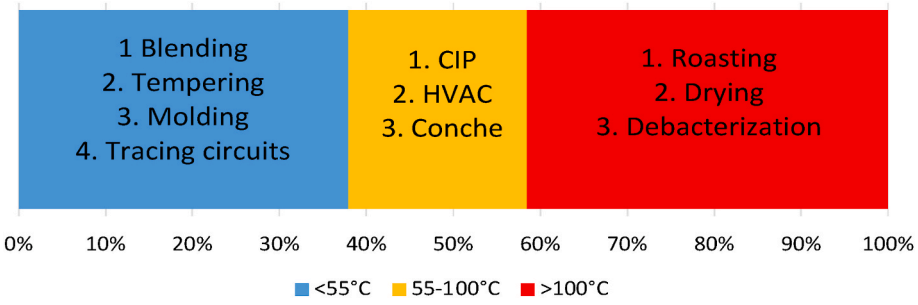


Fig. 4. Share of heat consumption by temperature levels in chocolate production.

Table 1
General categories of heat pumps based on the maximum achievable T_{sink} (Arpagaus et al., 2018).

Heat pump category	Maximum T_{sink} (°C)
Standard heat pump (SHP)	80
High temperature heat pump (HTHP)	80–100
Very high temperature heat pump (VHTHP)	100–160

production of the companies for which the pinch studies are available and are presented in Table 2 along with the typical process stream data.

Companies in the Swiss chocolate industry are segmented into four categories based on the estimates of annual production. Typical company sizes are selected based on the most commonly occurring annual production estimates within a certain category (see Table 3). For example, the majority of companies belonging to "very small" company category have annual production of approximately 1,000 tonnes. The annual production for each company is estimated using the average annual production per employee obtained from the statistics published by ChocoSuisse (2021)and the number of employees per company obtained from the STATENT database (FSO, 2023). Based on the estimates of total electricity and fuel consumption, the electricity and natural gas prices paid by each company are identified using the categories defined by the Elcom (ElCom, 2023) and the survey of natural gas prices (DFE and L'ÉCONOMIE, 2023). Cost-effectiveness of various commercially available HPs in the European market is analyzed for these typical company sizes and for energy prices paid by the company categories.

2.2. Techno-economic data for heat pumps

The technical data (such as typical temperature ranges, heating/cooling capacities, working fluids and compressor types) and corresponding cost data (such as HP cost, installation cost and annual operation and maintenance costs for commercially available SHPs and HTHPs) for selected HP models are obtained from various literature sources and reports. In total, 30 HPs installed between 2008 and 2019 across Switzerland (Cordin Arpagaus, 2020), Germany (Schlosser et al., 2020), Austria, France and Denmark (Rainer and Jakobs, 2020) were identified from the aforementioned sources for which the technical and economic data along with operating conditions were reported (summarized in Table 4). The cost data from countries other than Switzerland are converted to Swiss Francs (CHF) using the purchasing power parity (PPP) exchange rates. Despite the growth in the market, HP costs reported in the literature have remained relatively constant for the past couple of decades (Schlosser et al., 2020). As a result, the HP costs are converted to 2021 CHF by only correcting for inflation using producer price indices (FSO). The required heating capacities of HPs for various sizes of the companies in the Swiss chocolate sector are estimated using the representative profiles for chocolate production (see 3.1 below).

Based on the cost data adopted for the Swiss market, a cost function is derived to estimate the specific investment cost of HP (CHF/kW heating capacity) as a function of the heating capacity for the HP (see Fig. 5). The specific costs for the HPs decrease exponentially and then levels off with the increase in the heating capacity. Specific investments for the HPs shortlisted in this study range between 2600 CHF/kW heating and 200 CHF/kW heating of heating capacity based on the adjusted costs. The cost function estimated based on these values is later used to assess the cost-effectiveness of HP integration for various size categories. However, it should be noted that the as a result of wide distribution of HP costs, the results of cost-effectiveness are presented as a range of levelized costs for saved energy and CO_2 emissions.

2.3. Energy prices and CO_2 levy

The annual energy efficiency (EE) improvement-related benefits generated by HP integration are estimated as the annual energy costs saved due to achieved net TFE savings, thereby using the energy prices for the year 2021. The average prices paid for natural gas within Switzerland are monitored for nine standardized consumer categories including three categories for industrial companies and are obtained from the survey of natural gas prices (DFE and L'ÉCONOMIE, 2023).[3] By analogy, the prices paid for electricity by the Swiss industry are also reported for four industrial consumer categories and obtained from Elcom (ElCom, 2023). The cost-effectiveness of HP options is assessed for three different ratios of electricity to natural gas prices typically paid by the companies in Switzerland (henceforth referred to as energy price ratio); (1) minimum energy price ratio (i.e., lower electricity price compared to natural gas), (2) average energy price ratio (i.e., the ratio of average electricity and natural gas price across cantons) and (3) maximum energy price ratio (i.e., higher electricity prices compared to natural gas prices). The annual benefits from fuel-saving measures also depend on the cost savings generated by CO_2 abatement. Annual CO_2 abatement-related cost savings are estimated using the CO_2 abatement generated through HP integration and the CO_2 levy. The CO_2 emission factors for natural gas (0.2 kg CO_2/kWh) is obtained from the Swiss national inventory report (FOEN, 2023) and the CO_2 emission factor for electricity generation is assumed to be 0.12 kg CO_2/kWh (Krebs and Frischknecht, 2021). The current CO_2 levy of 120 CHF/tonne CO_2 is used for the cost-benefit analysis in this study ((SFOE), 2023).

2.4. Discount rates and lifetime

Discount rates (DR) are used in the cash flow analysis to amortize the initial capital investment cost over the lifetime of the investment. DR used by the industrial decision makers to analyze the cost-effectiveness of EE-related investment vary based on the various techno-economic

[3] Natural gas prices are reported inclusive of CO_2 levy. However, in order to analyze the effect of net CO_2 savings, natural gas prices exclusive of CO_2 levy are estimated and economic benefits of CO_2 abatement are analyzed separately. For detailed prices and ratios of electricity to natural gas prices, refer to appendix 2.

Table 2
Average specific final energy consumption and typical process requirements in the Swiss chocolate sector.

Process	Production route*	Specific final energy consumption (kWh/tonne)	Process streams	Tin (°C)	Tout (°C)	Cp (KJ/Kg.K)	Modeling approach
Roasting (beans)	PR1	110	Roasting air in	20	148	1.890	GB
			Roasting air out	100	20	1.890	GB
Roasting (nibs)	PR2	133	Roasting air	25	110	1.80	GB
			Roasting air out	450	25	1.80	GB
IR drying	PR2	85	Exhaust Air	300	30	1.39	GB
Debacterization	PR1 & PR2	88	Air in	25	80	1.890	GB
CIP	PR1 & PR2	30	Clean water in	10	90	4.185	GB
Tracing circuits	PR1 & PR2	240	Warm water	25	35	4.189	GB
Conching	PR1 & PR2	60	Cooling water out	85	25	4.185	GB
Butter melting	PR1 & PR2	11	Cocoa butter in	43	50	1.950	WB
			Butter cooling	45	20	1.750	WB
Tempering	PR1 & PR2	110	Chocolate mass in	45	46	0.455	GB
Molding	PR1 & PR2	37	Liquid chocolate in	57	58	0.455	GB
HVAC	PR1 & PR2	69	Warm water	35	55	4.18	GB
Grinding	PR1 & PR2	30	Cooling water	25	95	4.18	GB
Refrigeration	PR1 & PR2	570	Industrial water (Desuperheater)	60	25	4.185	GB
			Cooling water	35	25	4.185	GB

*PR1) chocolate production via whole beans roasting and PR2) chocolate production via infrared (IR) drying and cocoa nibs roasting.

Table 3
Categories of chocolate manufacturing companies in Switzerland.

Company category	Size (annual production in tonnes)	Energy price category
Category 1 (Very small)	1,000	C3
Category 2 (Small)	10,000	C4
Category 3 (Medium)	20,000	C5, C6
Category 4 (Large)	30,000	C7

characteristics of the industrial companies. The DR used in the cash flow analysis by industry decision-makers typically reflect both the time value of money and the perceived risk for the investment decisions. In order to reflect the company-level investment decision criterion for the Swiss industry, a DR of 21% is assumed in the present analysis for cost-effectiveness analysis of HP integration based on the authors' previous work (Bhadbhade and Patel, 2024). A technological lifetime of the HP is assumed to be 20 years based on the typical technological lifetime of its components (Jibran et al., 2022).[4]

2.5. Exemplar profiles, representative profiles and sectorial profiles

As established before, the appropriate sizing and placement of energy conversion technologies such as HPs require accurate quantification of excess heat available from the process (source), process heat demand (sink), and temperature profiles. The methodology presented by Ong et al. (2023) is extended in this work by introducing the generation of exemplar profiles and presented graphically in Fig. 6. The exemplar profiles provide specific heating and cooling requirements for one tonne of product (kWh/$t_{product}$). These product-specific requirements are obtained by carrying a mass balance on each process stream using the data such as annual production or process equipment capacities reported in the previously carried Pinch Analyses. The exemplar profiles thus obtained provide flexibility to scale up the process requirements to represent the given production capacity (in tonnes) and operating case (e.g., summer, winter, transition period using operational hours). In current work, representative profiles for four size categories and two production routes are generated for the Swiss chocolate production. The representative profiles are presented in the form of GCC and provide information about the thermal power requirements (kW) for each size category. A representative net heat sink and source profile is created for each category (size and production route) of the company by scaling up[6] the net heat sources and sinks to the annual production and operating hours represented by all companies of the given category (equation (1)). For example, a representative profile for a company with an annual production of 1,000 t p.a. (i.e., "very small" companies) is created and scaled up to total annual production represented by all companies in this category. A profile for the entire chocolate production (i.e., product-specific energy demand profile) is created by aggregating the net heating and cooling demand in each temperature interval for all company categories obtained in the previous step. The various product-specific energy demand profiles can be aggregated to obtain sectorial profiles, providing energy demand profiles across the entire sector (e.g., food and beverage).

$$SF = \frac{TAP_{Sector}}{TAP_{Analysed\ Plant}} \quad \text{Eq. 1}$$

Where,

SF = Scaling factor.

TAP = Total annual production.

Based on the type of commonly employed technologies in chocolate production, two representative profiles are identified for two production routes in Swiss chocolate manufacturing: 1) chocolate production via whole beans roasting and 2) chocolate production via infrared (IR) drying and cocoa nibs roasting. Configuration of process steps and process conditions for the process steps other than roasting and drying remain unchanged for both production routes. However, since the HP integration for nibs roasting was found to be economically unattractive (section 3), and since all other process steps are similar for both production routes, only production route 1 is analyzed in detail in the present work.

[4] Although the technical lifetime for HP is assumed to be 20 years based on literature, in some cases it was found to be 15 years.

[5] Aggregated profiles developed for one product are termed product-specific energy demand profiles (current work), whereas profiles aggregated at the level of the entire sector are termed sectorial profiles.

[6] Scaling-up could also be performed by comparing the TFE demand of the studied company and the TFE demand of the sector. In this study, scaling is instead performed via physical production output. The scaling factors used in this study are presented in Appendix 4.

Table 4
Select commercially available heat pumps in European market (Germany (Schlosser et al., 2020), Switzerland (Arpagaus et al., 2018), Rest of EU (Rainer and Jakobs, 2020)).

Heat pump model	Working fluid	Compressor	Heating capacity (kW)	Maximum T_{sink} (°C)	Investment cost (2022 CHF)	Installation cost (2022 CHF)	Cost type[a]
SHP1	R134a	NA	20	80	44,709	58,121	HP
SHP2	R407c	Piston	61	90	162,255	243,383	Total
SHP3	R404A	Piston	64	100	106,297	138,187	HP
SHP4	R134a	Screw	77	80	47,690	61,997	HP
SHP6	R134A	NA	194	80	407,745	611,617	Total
SHP7	Ammonia	Gas driven	194	150	93,520	140,280	Total
SHP8	R134A	Piston	194	80	156,430	203,359	HP
HTHP1	R134A	Piston	200	80	293,388	381,404	HP
SHP9	R134A	Screw	220	80	118,677	154,280	HP
SHP10	R404A	Piston	240	100	159,217	238,826	Total
SHP11	R134A	Screw	260	80	155,781	202,515	HP
SHP12	R134A	Piston	274	80	417,859	626,788	Total
SHP13	R1234ze	Piston	276	120	139,582	181,457	HP
SHP14	Ammonia	Piston	370	150	261,100	391,649	Total
SHP15	R134A	Piston	382	80	195,703	254,414	HP
SHP16	Ammonia	Screw	410	150	157,613	236,420	Total
SHP17	R134A	Screw	410	80	146,777	190,811	HP
SHP18	R134A	Piston	420	80	359,897	539,846	Total
SHP19	R1234ze	Screw	471	120	161,288	350,000	HP
SHP20	Ammonia	Piston	507	90	247,983	322,378	HP
HTHP2	R1234ze	Screw	520	120	258,162	335,610	HP
SHP21	R134A	Screw	550	80	249,006	323,708	HP
SHP22	R134A	Screw	692	80	316,160	474,239	Total
HTHP3	R744	Screw	800	90	391,024	508,332	HP
SHP23	R134A	Screw	800	80	686,041	1,029,062	Total
SHP24	Ammonia	Piston	1000	150	347,562	451,831	HP
HTHP4	Ammonia	Piston	1050	150	205,488	267,135	HP
HTHP5	Ammonia	Piston	3000	150	3,605,996	5,408,994	Total
SHP25	Ammonia	Screw	3250	150	1,276,958	1,915,438	Total

[a] The costs are reported either only for "HP" or "Total" cost (i.e., HP cost + installation). Cases where only "Total" costs are reported, the HP costs are estimated assuming HP costs contribute to about 40% of total costs (Henrik Pieper et al., 2018).

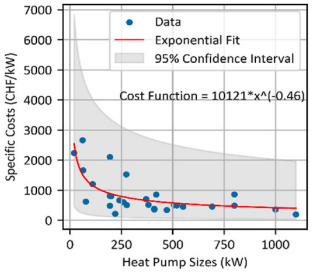


Fig. 5. Specific investment costs (without installation costs) for heat pumps as a function of heating capacity.

2.6. Selection of heat pumps

Commercially available HPs are shortlisted for the current study based on their application (i.e., for excess heat recovery) and representative profiles are used to identify the appropriate HP capacities for the given company size. The operating conditions, such as the temperature levels over which the HP is operated, and the temperature lift also affects the HP selection. The achievable temperature levels and temperature difference depend on the selection of working fluids, compressors, and the number of stages (Van de bor and Infante Ferreira, 2013). HPs that operate across operating conditions as identified for the Swiss chocolate industry are selected for this study. The required thermal power determined from the representative profiles are also taken into account while identifying HPs for each company size.

2.6.1. Coefficient of performance for the heat pump

The performance of HP is categorized using the COP. The theoretical maximum achievable COP_{carnot} for the HP is calculated by equation (2). The COPs achieved in a practical setting are, however, lower than the COP_{carnot} and represented as the ratio of the amount of useful heat (Q_{cond}) at T_{sink} to the electric power consumed by the compressor (W_{in}). The ratio of practical COP_{real} with theoretical COP_{carnot} provides the second law efficiency (η_{2nd}) of the HP system. Operating conditions and reported COPs provided for already implemented HPs are used to estimate η_{2nd} the HPs in the given operating temperature ranges (using equation (4)). The values of η_{2nd} thus estimated are then used to estimate the COP_{real} values for all HPs included in this work over the operating conditions identified from the representative profiles for chocolate production in Switzerland.

$$COP_{carnot} = \frac{T_{cond,HP}}{\Delta T_{lift}} \quad \text{Eq. 2}$$

$$COP_{real} = \frac{Q_{cond}}{W_{in}} \quad \text{Eq. 3}$$

$$\eta_{2nd} = \frac{COP_{real}}{COP_{carnot}} \quad \text{Eq.4}$$

COP_{real} for HPs are then compared with the threshold COPs for the companies of all categories and different ratios of electricity to natural gas prices to identify HPs with COPs in an economically feasible range.

2.6.2. Optimal heat pump integration

The representative profiles explained in the previous section allow to identify various possible T_{sink} and T_{source} temperatures, the required heat load above the pinch and available excess heat below the pinch for all types of companies. The HP should be integrated to operate across the pinch point. The COP curves in combination with GCC help to estimate the optimum condenser and evaporator duties along with the evaporator integration point. Corresponding condensation and evaporation

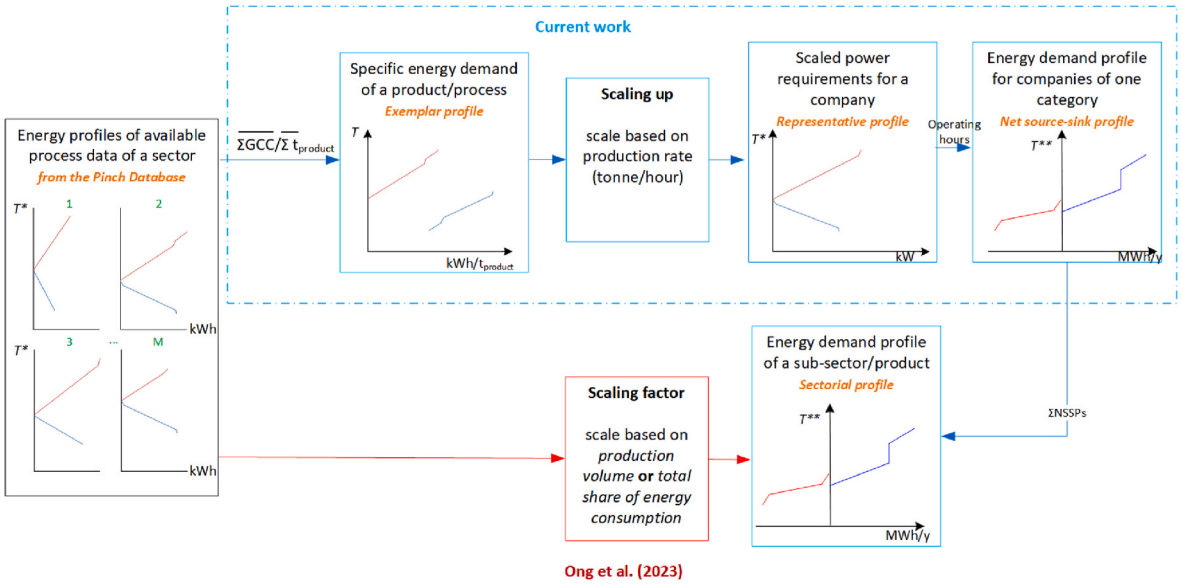


Fig. 6. Methodology for exemplar, representative and sectorial profiles.[5]

temperatures for HP are obtained based on the minimum temperature difference for heat exchangers.[7] If the amount of excess heat available at T_{source} is more than the amount of required process heat at T_{sink}, a COP curve (evaporator duty as a function of evaporation temperature is plotted by estimating the evaporation capacities for various evaporation temperatures in the range from pinch point to coldest T_{source}. Evaporator duty as a function of evaporation temperature is estimated using equations (2)–(5). If the amount of excess heat available is smaller than the required heat load, the thermodynamically optimum condenser load is estimated by rearranging equation (5). The point of intersection of COP curve with the GCC provides the optimum integration point and evaporator duty of the HP for the given T_{sink}.

$$Q_{evap,HP} = Q_{cond,HP} \times \left(1 - \frac{1}{COP_{real}}\right) \quad \text{Eq. 5}$$

2.7. Net energy savings

HP integration results in the reduction of fossil fuel consumption and an increase in electricity consumption. The thermal power required by the HP on the evaporator side depends on its COP_{real} (see eq. (3)). COP_{real} for commercially available HPs identified for the given operating conditions are estimated as explained in section 2.6.1. Annual electricity consumption is then estimated using the thermal power of the HP and annual operating hours. Full load operating hours for the HP are assumed based on the typical annual operating hours of the processes identified as sinks and sources. Annual net TFE savings are estimated as the difference between fuel saved, and electricity consumed.

2.8. Cost-effectiveness analysis

The Levelized Cost of Saved total final Energy (LCSE) and the Levelized Cost of CO_2 abatement ($LCCO_2$) are estimated to assess the economic attractiveness of HPs in the Swiss chocolate sector (see eq. (6)) for the given source and sink temperatures. The levelized cost for each investment in HP is reported such that negative values indicate profitability, while positive values imply higher costs. Levelized costs are used to identify the threshold COPs for HP integration and to estimate the overall cost-effective EE improvement potential for the entire chocolate sector.[8]

$$LCSE = \frac{I \times ANF + O\&M - B}{ES}, LCCO2 = \frac{I \times ANF + O\&M - B}{CA} \quad \text{Eq. 6}$$

Where,

I= Initial investment cost for HP
ANF = Annuity factor for the given discount rate and technology lifetime
O&M = Operation and maintenance cost
B = Annual benefits, i.e., cost of saved total final energy and of avoided CO_2 tax payments
ES = Total final energy savings
CA= CO_2 abatement potential

3. Results and discussion

Following the methodology explained in section 2, this section presents the identified temperature levels and sizes of heat sinks and sources for the Swiss chocolate sector, economic feasibility analysis for generic HP integration and cost-effectiveness analysis for integration of commercialized HPs.

3.1. Heat source and sink identification and heat pump dimensioning

Typical thermal power requirements for various process steps are estimated based on SECs, pinch reports and manufacturer catalogs for various equipment. Table 5 presents typical power requirements of the selected process steps for the company of categories 1 to 4 in the Swiss chocolate sector. The required power is highest for refrigeration, followed by hot water generation, roasting, conching and CIP. The excess heat available from refrigeration re-cooling is estimated using the energy efficiency ratio for the ammonia chiller obtained from (Seck et al., 2013), and available excess heat from conches is estimated based on the energy balance over the conches. However, it should be noted that the

[7] The condenser temperature is estimated using equation and evaporator temperature is estimated using equation STAMPFLI, J. A., ATKINS, M. J., OLSEN, D. G., WELLIG, B., WALMSLEY, M. R. W. & NEALE, J. R. 2018. Industrial heat pump integration in non-continuous processes using thermal energy storages as utility-A graphical approach. Chemical Engineering Transactions, 70, 901–906.

[8] For details of cost-effectiveness methodology refer to appendix 1.

Table 5
Typical power requirements* for selected process steps in chocolate manufacturing.

Process	Required power (kW) for size category 1 (1000 tonne/year)	Required power (kW) for size category 2 (10,000 tonne/year)	Required power (kW) for size category 3 (20,000 tonne/year)	Required power (kW) for size category 4(30,000 tonne/year)
Roasting	30	230	500	1000
Hot water generation	50	400	800	1,200
CIP	6	60	120	240
Conching	16	150	300	500
Refrigeration	110	2,200	2,500	4,600

* The power requirements provided in the table represent average values. They typically vary over a range of approx. ±10% for each size category depending on the type of chocolate produced.

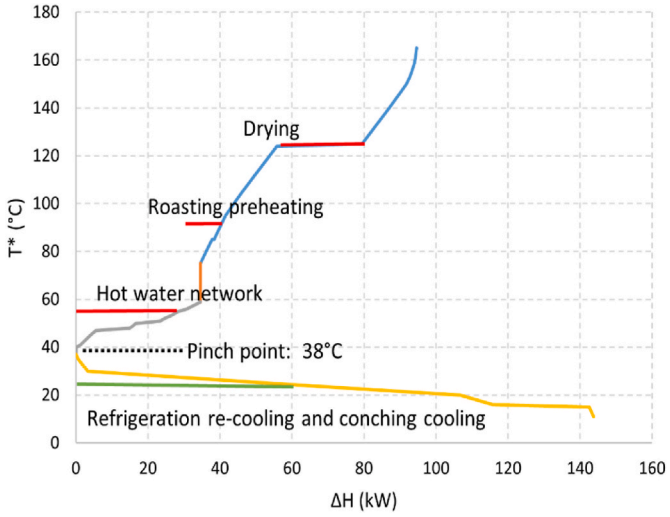


Fig. 7. Representative grand composite curves (profiles) for chocolate production in Switzerland.

typical power requirements are estimated by assuming the uniform SEC across all company sizes. As a result of larger companies running processes for extended durations, the increase in the thermal power with the company size is not linear.

Fig. 7 presents the representative profile for the Swiss chocolate industry. Two sources of excess heat (green line in Fig. 7) and three heat sinks (red lines in Fig. 7) separated by the pinch point at 38 °C are identified from the representative profile. The condenser of a refrigeration unit using ammonia as working fluid is cooled using cooling water, which typically enters the plant at 10 °C. Cooling water recovers the heat rejected from the condenser at 35 °C and typically leaves the plant at ambient temperature (25 °C assumed). Thus, the excess heat from the re-cooling of the refrigeration unit condenser can be recovered in the temperature range between 35 °C and 25 °C. Similarly, the excess heat available from conching cooling water can be recovered in the temperature range between 55 °C and 25 °C. A significant amount of heat is required for warm water production at 55 °C, which is used in the tracing circuits to keep the chocolate in molten form. Convection roasting of whole cocoa beans is carried out with hot air. Excess heat from refrigeration re-cooling or conching cooling can be utilized as a heat source for a HTHP enabling preheating the roasting air up to 90 °C. Heating required for CIP at 50 °C can also be provided by recovering excess heat with a HP from the aforementioned excess heat sources in a typical chocolate manufacturing company. The HP integration for roasting air preheating competes with the internal direct heat recovery from roaster exhaust air (without HP). The representative profile in Fig. 7 is representative of a company of size category 1 and included to illustrate the typical temperature levels identified for heat sinks and sources. While the required HP capacities for other size categories are presented in Table 6, the representative profiles for other category sizes and production routes are included in the appendix 3.

Table 7 below presents the required heating demands (condenser duty) and the evaporator duty for the various HP integration opportunities identified above based on the representative profile for the Swiss chocolate manufacturing companies belonging to the categories 1 to 4. The evaporator load required for the identified condenser duty is estimated using the COP curves (refer to appendix 3 for GCC-specific COP curves). The COP curves are estimated based on the COP_{real} of the most widely used working fluids. The COP_{real} for each working fluid is obtained based on the average second law efficiencies for HPs with a particular working fluid, as explained in section 2.6. According to the representative profile for the chocolate factory, the sink temperature can be identified at 55 °C. The required condenser duty for the HP (i.e., heating capacity) integration varies between 40 kW for the company of the size category 1–1200 kW for the company of the size category 4. Based on the COP curves for HPs with various working fluids, the optimum temperature for evaporator placement for the ammonia HP is 23 °C, for R1234 ze is 25 °C and R134a HPs is 27 °C. The HP evaporator size varies from 20 kW for category 1 company to for to 800 kW for the company of category 4.

3.2. Bottom-up sector wide heat pump integration potential

Following the methodology presented in section 2.5, the product-specific energy demand profile for the Swiss chocolate manufacturing is presented in Fig. 8. Based on the profile for the entire Swiss chocolate production, there is a total requirement of 68,807 MWh of heat at 55 °C to produce hot water (AR1/AR3) and 15,000 MWh at 90 °C for preheating the roasting air (AR2/AR4) across all companies in the Swiss chocolate industry. Together these two opportunities represent 23% TFE reduction potential and 45% CO_2 emissions reduction potential from the Swiss chocolate industry (see Table 8). Drying process in the chocolate industry typically consumes direct steam at 120 °C. Replacing the direct steam supply with steam generating HP represents an additional sector-wide potential of about 12,000 MWh. This steam demand can be further replaced using the steam generating HP increasing the total identified potential to 28% for TFE reduction and 56% for the CO_2 abatement (see Table 8). However, due to the lack of cost data related to steam generating HPs, their cost-effectiveness is not assessed in this work. The excess heat available from the refrigeration re-cooling for the entire sector is estimated at 95,000 MWh across companies of all sizes and production routes. However, based on the estimates of the evaporator duties, only 54,000MWh of excess heat is required to realize the available HP integration potential in the Swiss chocolate industry. This implies the remaining 41,000 MWh excess heat available from the Swiss chocolate manufacturing companies can be used for other purposes such as space heating or supply to district heating networks. As a caveat, it should also

[9] Data for product-specific energy demand for Swiss chocolate production can be found in appendix 6.

Table 6
Theoretical heat pump integration opportunities identified for Swiss chocolate manufacturing.

HP arrangement	Source	Sink	T_{source} (°C)	T_{sink} (°C)	Temperature lift ΔT_{lift} (°C)	HP type
AR1	Refrigeration re-cooling	Warm water network (tracing circuits), CIP	25	55	30	SHP
AR2	Refrigeration re-cooling	Roasting pre-heating	25	90	65	HTHP
AR3	Conching cooling water	Warm water network (tracing circuits), CIP	23	55	27	SHP
AR4	Conching cooling water	Roasting pre-heating	23	90	67	HTHP

be noted that the estimates of sector-wide potential are based on publicly available information about HP implementation in the Swiss chocolate industry, possibly resulting in double counting for some companies which have already implemented HPs. This may lead to some overestimation of the remaining sector-wide HP integration potential for the Swiss chocolate industry.

3.3. Economic feasibility of heat pump configuration for Swiss chocolate industry

The cost-effectiveness of the HP integration is estimated using the LCSE, which depends on the cost of HP, the COP of the HP, and energy prices paid by the company. The COP of the HP, in turn, depends on the temperature lift, operating hours, and the second law efficiency of the HP. The economic feasibility of various possible HP arrangements identified in section 3.1 is assessed using a threshold COP. The LCSE for various possible HP configurations is estimated for different COP values. A COP value at which the HP investment becomes cost-effective is termed threshold COP. Any HP that can operate with a COP above the threshold COP will result in a profitable investment for the given HP arrangements and the associated assumptions about HP cost, operating hours and average energy prices. The HP costs are estimated for the given size of the heat sink using the cost function presented in Fig. 5. The typical operating hours for the processes considered for the heat sinks and sources are estimated based on the information available from the pinch studies (Ong et al., 2023). Swiss industry average electricity and natural gas prices are assumed for the economic feasibility assessment by company size. A typical refrigeration plant in a chocolate manufacturing facility runs year-round at approximately constant power. The heat is required continuously throughout the year for hot water production used in the tracing circuits. As a result, the HP arrangement AR1 (i.e., HP integration between refrigeration re-cooling and tracing circuits) has the lowest threshold COPs among the identified HP configurations, ranging from 2.8 (very small company) to 1 (large company). In other words, any HP that can operate with the COP_{real} of 2.8 and above can be integrated cost-effectively for HP configuration AR1 in a company of any size category in the Swiss chocolate industry. Since the HPs with smaller heating capacities have relatively higher specific costs and smaller companies pay relatively higher energy prices, the threshold COPs for size category 1 are considerably higher compared to the larger size categories. Conching process, on the other hand, is typically operated in batches, making it an intermittent heat source for HP configuration AR3 and technically less viable (i.e., HP integration between conching cooling and tracing circuits). As a result, the threshold COPs for AR3 range between 6 for small companies to 1.4 for large companies. A threshold COP value of at least 6 makes many of the currently commercially available HP options economically unviable for AR3 in a company of size category 1. Threshold COPs for HP arrangement AR3 in other company sizes are also higher compared to AR1 for the same size category. Similarly, for AR2 and AR4, with the operating hours being limited by the availability of the heat sink (i.e., roasting preheating), any HP with a COP_{real} value below 6.4 cannot be integrated cost-effectively (see Fig. 9) for the average energy prices in Swiss industry and estimated HP cost. Since the COP_{carnot} for the ΔT_{lift} required for AR2 and AR4 is 6, the HP integration for both of these configurations in the Swiss chocolate industry is non-cost-effective. Moreover, based on the literature reviewed, most cocoa roasters employed in chocolate manufacturing already make use of direct internal heat recovery and direct internal heat recovery is often more cost-effective than HP integration for heat requirements at high temperatures (Bhadbhade and Patel, submitted). It should, however, be noted that the Levelized costs of saved final energy are estimated assuming the more conservative discount rate of 21% to reflect the company-level decision-making criteria in Swiss industry, whereas typical private discount rates used for discounted cash flow analyses in the literature range between 8% and 10%.

3.4. Cost-effectiveness analysis for commercially available heat pumps in the Swiss chocolate industry

The following section presents the range of levelized costs of saved final energy and the CO_2 abatement for commercially available HP options from the European market for the operating conditions identified in the Swiss chocolate industry. The cost-effectiveness is assessed for three different ratios of electricity to natural gas prices for the Swiss chocolate industry (see section 2.3). Based on the selection criteria explained in section 2.6, four commercially available SHPs on the European market can be potentially integrated in the company of size category 1. Relatively higher specific investment costs for HPs with smaller heating capacities in combination with a relatively higher energy price ratio on average for the company of size category 1 results in the majority of commercially available SHPs appearing non-cost-effective for the companies of category 1 for the average energy price ratio and none of the commercially available HPs appear cost-effective for maximum energy price ratio. Integration of HPs for companies paying average or high energy prices can cost in the range of 300–600 CHF/t-CO_2 abatement (or 0.3 to 0.6 CHF/kWh of saved final energy). However, for the companies paying the minimum energy prices, all commercially available HPs for size category 1 can be integrated cost-effectively (see Fig. 10).

The representative profile for the company of the size category 2 indicates an HP heating capacity of 400 kW for the arrangements AR1 or AR3 and HTHP of the capacity 200 kW for the HP arrangements AR2 or AR4 (see Table 7 and appendix 3). Based on the selection criteria (see section 2.6), there are 17 commercially available SHP options, which can be used to valorize the excess heat available from the refrigeration re-cooler and use it for hot water production for tracing circuits and 3 HTHP options for supplying heat for the roasting preheating. Due to the lower energy price ratio for the company of size category 2 compared to the size category 1 and relatively lower specific costs for HPs required for the identified heating capacities, the majority of the SHP options appear cost-effective and more profitable than size category 1. However, for the average and maximum price ratios, some of the commercially available HTHP options can cost as much as 200 to 400 CHF/t-CO_2 abated (or 0.1 to 0.2 CHF/kWh of saved final energy). All commercially available HP options appear economically viable for companies falling into category 2 and paying minimum energy prices (see Fig. 11).

The representative profile of the company category 3 indicates the required condenser duty for the HP to be 800 kW for the HP arrangements AR1 or AR3 and 400 kW for the arrangements AR2 or AR4. Based on HP selection criteria, there are two commercially available SHP options and one HTHP option for the company of size category 3. Since the reported COPs for the given operating conditions are higher than 1.2 and due to the relatively low average energy price ratio, all SHP options are

Table 7
Estimated condenser capacities and evaporator capacities for various heat pump arrangements and company sizes.

	Size category 1		Size category 2		Size category 3		Size category 4	
HP arrangement	Condenser duty (kW)	Evaporator duty (kW)	Condenser duty (kW)	Evaporator duty (kW)	Condenser duty (kW)	Evaporator duty (kW)	Condenser duty (kW)	Evaporator duty (kW)
AR1	40	31	400	280	800	645	1,200	850
AR2	20	12	200	114	400	228	700	400
AR3	40	31	400	280	800	645	1,200	850
AR4	20	12	200	114	400	228	700	400

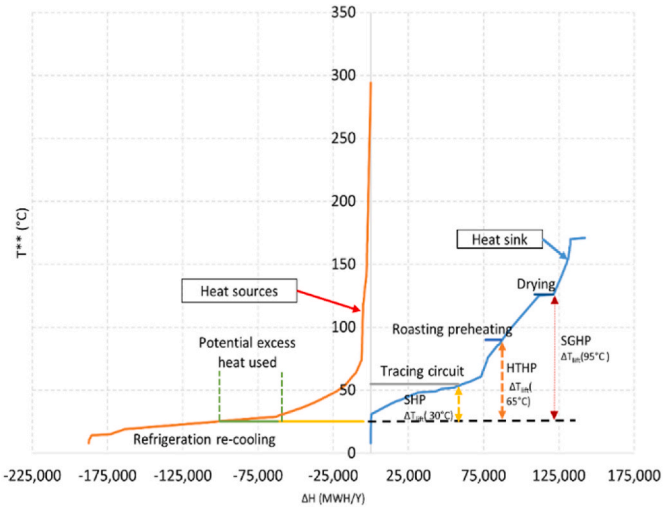


Fig. 8. Product-specific energy demand profile for Swiss chocolate sector with theoretical heat pump integration potential (*Potential excess heat utilized: Excess heat that can potentially be utilized for HP integration opportunities*).[9]

Table 8
Sector-wide technical TFE reduction and CO_2 abatement potentials for various heat pump arrangements.

HP arrangement	HP type	Sector-wide potentials	
		TFE reduction	CO_2 abatement
AR1/AR3	SHP	18%	34%
AR2/AR4	HTHP	5%	11%
Steam for drying process	Steam generating heat pump	5%	11%
Total		**28%**	**56%**

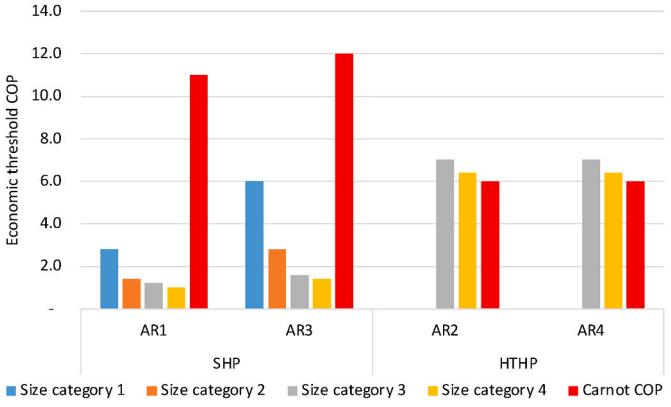


Fig. 9. Threshold COPs for different heat pump integration configurations in the Swiss chocolate industry.

economically attractive for the company of category 3 for the minimum and average price ratio. The HTHP options become economically viable only for the companies paying minimum energy prices (see Fig. 12). For the companies paying average and maximum prices, the HTHP integration opportunities can cost 50 to 200 CHF/t-CO_2 abatement.

Finally, the company of size category 4 requires SHP of the size 1200 kW to valorize the heat from the refrigeration re-cooler and use it for hot water production and 600 kW HTHP for the heat sink of roasting preheating. Two SHP options and one HTHP are identified for the company of size category 4 and their levelized costs and potentials are presented in Fig. 13. Similar to companies from size categories 2 and 3, both

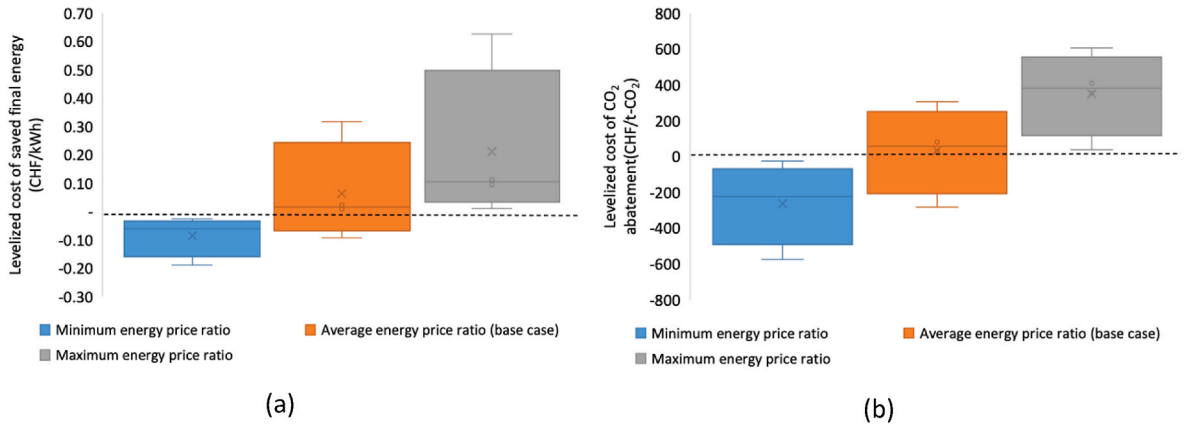


Fig. 10. (a) Levelized costs of saved final energy for commercial HP options for size category 1 (b) Levelized cost of CO_2 abatement for commercial HP options for size category 1.

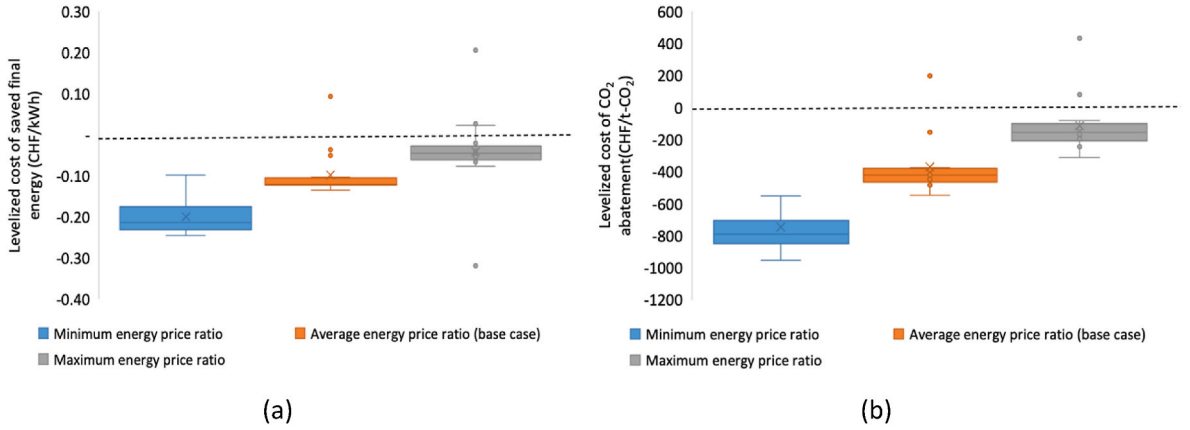


Fig. 11. (a) Levelized costs of saved final energy for commercial HP options for size category 2 (b) Levelized cost of CO_2 abatement for commercial HP options for size category 2.

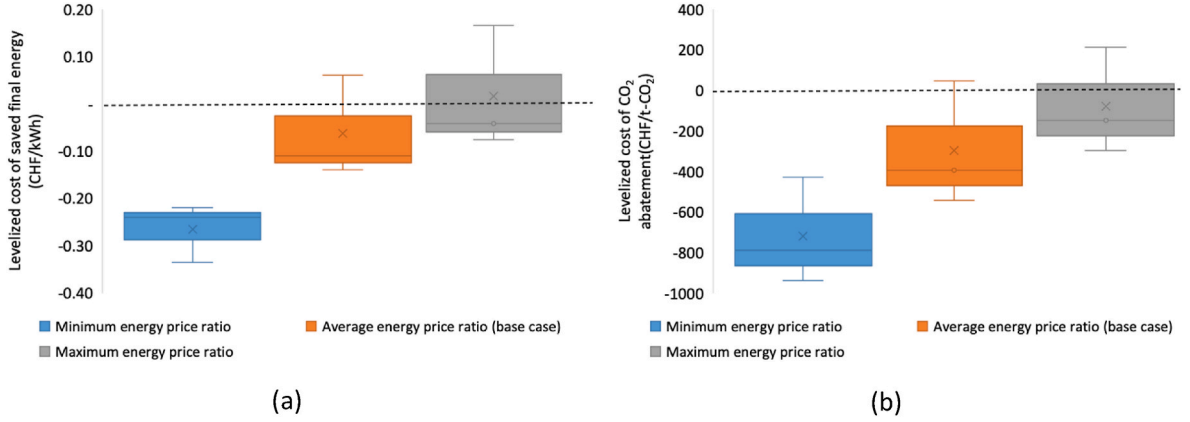


Fig. 12. (a) Levelized costs of saved final energy for commercial HP options for size category 3 (b) Levelized cost of CO2 abatement for commercial HP options for size category 3.

commercially available options of SHPs appear economically attractive for the companies paying average and minimum energy prices in size category 4. The HTHP option becomes economically viable only for companies paying minimum energy prices. For companies paying average or higher energy prices, the HTHP integration can cost in the range of 40–140 CHF/t-CO_2 abatement.

The cost-effectiveness of HPs is assessed based on the initial investment cost, which is obtained from the cost function created using the cost data provided in the literature. Although the literature sources used for data in this study offer many case studies of HP integration across Europe, the list is not exhaustive, and some commercially available HPs may be less (or more) expensive than the HPs included in the study. Moreover, only HPs for which the cost data is available are considered for the cost-effectiveness analysis, which resulted in the omission of some HP options with appropriate heating capacities and operational conditions identified for the company sizes. For an average chocolate

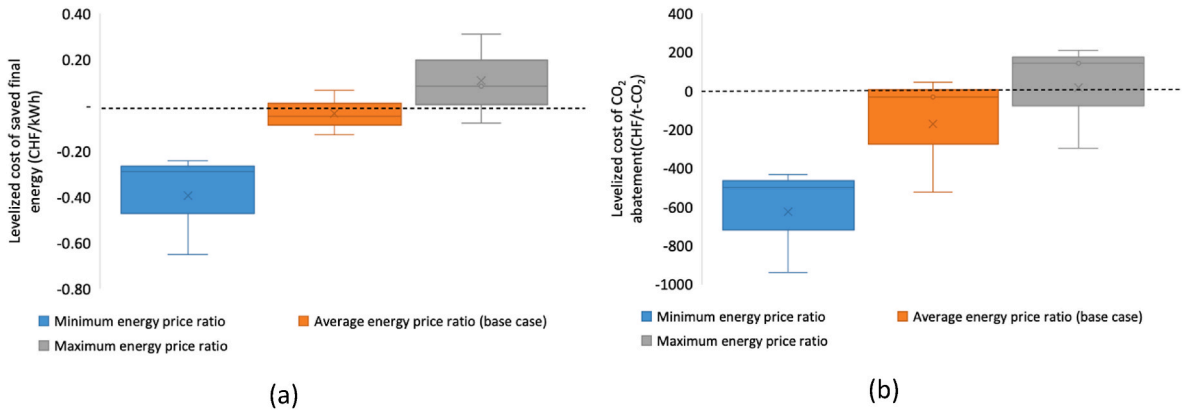


Fig. 13. (a) Levelized costs of saved final energy for commercial HP options for size category 4 (b) Levelized cost of CO_2 abatement for commercial HP options for size category 4.

manufacturing company, the quantity of excess heat available is considerably higher than the process heat demand. This heat can be used for more ambitious CO_2 emission reduction in the sector studied for space heating (direct use or via district heating). The techno-economic analysis of these solutions could be assessed in future work. Despite its limitations, the current work provides a framework to estimate the bottom-up sector-wide HP integration potential using representative profiles and sectorial profiles.

4. Conclusions

The current work presents the methodology to generate exemplar, representative and sector-wide profiles for energy requirements in the process industry and use these profiles to estimate the potential for heat pump (HP) integration. The methodology is presented using the Swiss chocolate industry as a case study. The data from previously conducted Pinch Analyses allows the generation of exemplar profiles for the given product. The exemplar profiles provide heating and cooling requirements for one tonne of product. The representative profiles can then be generated to represent the scale of a given production capacity, which can be used to identify different integration opportunities and required capacities for energy conversion technologies such as HPs. Identifying HP integration opportunities using representative profiles also facilitates cost-effectiveness analysis of HP integration opportunities for various company sizes. The data for individual companies are aggregated and scaled up using the annual production to create a profile of annual heating and cooling demands for more comprehensive scopes. Thus, it facilitates the estimation of the potentials for energy efficiency improvement and decarbonization using energy conversion technologies as well as industrial waste heat for different scopes such as products, product groups or industry sub-sectors.

Based on the application of the methodology outlined in this work, representative profiles for four company sizes are generated for Swiss chocolate production. Overall, five HP integration opportunities can be identified using these profiles. The company size and temperatures and operating schedules of sources and sinks affect the cost-effectiveness of HP solutions. Pinch temperatures at 38 °C, relatively lower ΔT_{lift}, and simultaneous heating and cooling requirements in most cases for chocolate production make the integration of most commercially available SHPs with real coefficients of performance above 2.6 technically feasible and profitable within companies of all sizes and most energy price ratios. However, the HTHP integration for the companies paying high electricity prices can cost up to 600 CHF/t-CO_2 abatement for the Swiss chocolate industry. It should be noted that despite using very stringent criteria for the cash flow analysis (identified based on previous research), most commercially available HP options appear cost-effective, thus implying HP integration to be a profitable investment even for the most risk-averse companies in the Swiss chocolate industry. Overall, HP integration in the Swiss chocolate production can potentially achieve up to a 28% reduction in TFE and a 56% reduction in annual CO_2 emissions. Despite its limitations and the focus on HP integration in chocolate production, the methodology presented in this work can be replicated to assess the integration potentials for other energy conversion technologies (e.g., solar thermal, CHP, etc.), different industrial sectors, and different geographic scopes.

CRediT authorship contribution statement

Navdeep Bhadbhade: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Benjamin H.Y. Ong:** Writing – review & editing, Methodology. **Donald G. Olsen:** Writing – review & editing. **Beat Wellig:** Writing – review & editing. **Martin K. Patel:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

None of the authors of this paper has a financial or personal relationship with other people or organizations that could inappropriately influence or bias the content of the paper.

Data availability

The data that has been used is confidential.

Acknowledgement

We acknowledge financial support by both the Chair for Energy Efficiency at University of Geneva and the project SWEET-DeCarbCH (Decarbonization of Cooling and Heating in Switzerland) funded by the Swiss Federal Office of Energy (SFOE), contract number Si/502260-01. The authors also gratefully acknowledge the exchange with Prof. François Maréchal (EPFL), ultimately allowing them access to the company data used in this analysis.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2024.142323.

References

Agner, R., Ong, B., Wellig, B., 2021. Integration of heat pumps and thermal energy storages in non-continuous industrial processes. Bundesamt für Energie BFE. Bern.
Arnold, O., Brühlmeier, H., 2010. Gesamtenergieanalyse mit der Pinch-Methode: Heineken Switzerland AG; Chur.

Arpagaus, C., 2019. Maestrani setzt auf Energieeffizienz. https://www.foodaktuell.ch/2019/10/03/maestrani-setzt-auf-energieeffizienz/.
Arpagaus, C., Bless, F., Uhlmann, M., Schiffmann, J., Bertsch, S.S., 2018. High temperature heat pumps: market overview, state of the art, research status, refrigerants, and application potentials. Energy 152, 985–1010.
Becker, H., Vuillermoz, A., Maréchal, F., 2012. Heat pump integration in a cheese factory. Appl. Therm. Eng. 43, 118–127.
BFE, 2023a. Energieverbrauch in der Industrie und im Dienstleistungssektor. Bundesamt für Energie (BFE).
BFE, 2023b. Energy Perspectives 2050+. https://www.bfe.admin.ch/bfe/en/home/policy/energy-perspectives-2050-plus.html/:SwissFederalOfficeofEnergy. SFOE.
Bhadbhade, N., Patel, M., 2020. Analysis of energy efficiency improvement and carbon dioxide abatement potentials for Swiss Food and Beverage sector Analysis of energy e ffi ciency improvement and carbon dioxide abatement potentials for Swiss Food and Beverage sector, p. 161.
Bhadbhade, N., Patel, M.K., 2024. Energy efficiency investment in Swiss industry: analysis of target agreements. Energy Rep. 11, 624–636.
Chocosuisse, 2021. Facts & figures [Online]. Available: https://www.chocosuisse.ch/en/services/facts-figures [Accessed 2.02.2023].
Cordin Arpagaus, S.B., 2020. Industrial heat pumps in Switzerland: application potentials and case studies. Swiss Federal Office of Energy (SFOE).
DFE, 2023. In: L'ÉCONOMIE, D.F.D. (Ed.), Surveillant des prix relatif aux prix du gaz naturel. https://gaspreise.preisueberwacher.ch/web/index.asp?l=1.
EEA, 2023. EEA Greenhouse Gases — Data Viewer.
ELCOM, 2023. Prix de l'électricité en Suisse. https://www.strompreis.elcom.admin.ch: ElCom.
ENERGIE-AGENTUR DER WIRTSCHAFT ENAW, 2016. Database of Energy Efficiency Measures.
Flükiger, U., W, T., 2011. ENERGIEANALYSE MIT DER PINCH-METHODE SÜSSWARENHERSTELLUNG. Swiss Federal Office of Energy (SFOE).
FOEN Switzerland's Greenhouse Gas Inventory 1990–2021: National Inventory Document and reporting tables (CRF). Submission of April 2023 under the United Nations Framework Convention on Climate Change, 2023. Federal Office for the Environment, Bern. http://www.climatereporting.ch:.
FSO, 2023. Structural Business Statistics (STATENT) from 2011 Onwards. https://www.bfs.admin.ch/bfs/en/home/services/geostat/swiss-federal-statistics-geodata/business-employment/structural-business-statistics-statent-from-2011-onwards.html.
Henrik Pieper, T.O., Buhler, Fabian, Lava Paaske, Bjarke, Elmegaard, Brian, Markussen, Wiebke Brix, 2018. Allocation of investment costs for large-scale heat pumps supplying district heating. CONECT 2018.
IEA, 2019. Renewables 2019. International Energy Agency (IEA). https://www.iea.org/reports/renewables-2019.
IEA, 2022. Electrification. IEA. https://www.iea.org/reports/electrification.
Jesper, M., Schlosser, F., Pag, F., Walmsley, T.G., Schmitt, B., Vajen, K., 2021. Large-scale heat pumps: uptake and performance modelling of market-available devices. Renew. Sustain. Energy Rev. 137.
Jibran, S., Zuberi, M., Hasanbeigi, A., Morrow, W., 2022. Bottom-up assessment of industrial heat pump applications in U.S. Food manufacturing. Energy Convers. Manag. 272.
Kantor, I., Santecchia, A., Wallerand, A. S., Kermani, M., Salame, S., Arias, S., Wolf, F. & Eetvelde, G. V. Thermal profile construction for energy-intensive industrial sectors. Proceedings Of Ecos 2018 - The 31st International Conference On Efficiency, Cost, Optimization, Simulation And Environmental Impact Of Energy Systems, 2018 Guimaraes, Portugal.
Kapustenko, P.O., Ulyev, L.M., Boldyryev, S.A., Garev, A.O., 2008. Integration of a heat pump into the heat supply system of a cheese production plant. Energy 33, 882–889.
Klemes, J., Dhole, V.R., Raissi, K., Perry, S.J., Puigjaner, L., 1997. Targeting and Design Methodology for Reduction of Fuel, Power and CO2 on Total Sites.
Krebs, L., Frischknecht, R., 2021. Umweltbilanz Strommixe Schweiz 2018. Uster.
Meier, D., 2006. Pilotphase Pinch-Methodik 2006/07 Projekt: Bell AG.
Meier, D., 2008. Pinch-Analyse 2008 Projekt: Lindt & Sprüngli AG. Olten: Swiss Federal Office of Energy (SFOE).
Olsen, D.G., Abdelouadoud, Y., Liem, P., Hoffmann, S., Wellig, B., 2017. Integration of heat pumps in industrial processes with pinch analysis. In: 12th IEA Heat Pump Conference. Rotterdam.
Ong, B.H.Y., Bhadbhade, N., Olsen, D.G., Wellig, B., 2023. Characterizing sector-wide thermal energy profiles for industrial sectors. Energy 282.
Rainer, M., Jakobs, C.S., 2020. IEA Annex 48: Industrial Heat Pumps, Second Phase. Germany: Heat Pump Centre.
Schlosser, F., Arpagaus, C., Walmsley, T.G., 2019. Heat pump integration by pinch analysis for industrial applications: a review. Chemical Engineering Transactions 76, 7–12.
Schlosser, F., Jesper, M., Vogelsang, J., Walmsley, T.G., Arpagaus, C., Hesselbach, J., 2020. Large-scale heat pumps: capabilities, feasibility and industrial integration. Renewable Sustainable Energy Rev., 110219 submitted for publication.
Seck, G.S., Guerassimoff, G., Maïzi, N., 2013. Heat recovery with heat pumps in non-energy intensive industry: a detailed bottom-up model analysis in the French food & drink industry. Appl. Energy 111, 489–504.
Seevers, J.-P., Schlosser, F., Peesel, R.H., Hesselbach, J., 2018. Dimensioning of heat pump systems based on pinch analysis and energy monitoring data. CHEMICAL ENGINEERING TRANSACTIONS.
(SFOE), S. F. O. O. E, 2023. *CO2 levy* [Online]. https://www.bafu.admin.ch/bafu/en/home/topics/climate/info-specialists/reduction-measures/co2-levy.html.
Stampfli, J.A., Atkins, M.J., Olsen, D.G., Walmsley, M.R.W., Wellig, B., 2019. Practical heat pump and storage integration into non-continuous processes: a hybrid approach utilizing insight based and nonlinear programming techniques. Energy 182, 236–253.
Townsend, D.W., Linhoff, B., 1983. Heat and power networks in process design: Part 1: criteria for placement of heat engines and heat pumps in process networks. AIChE J. 29, 742–748.
Van de bor, D.M., Infante Ferreira, C.A., 2013. Quick selection of industrial heat pump types including the impact of thermodynamic losses. Energy 53, 312–322.
Wallerand, A.S., Kermani, M., Kantor, I., Maréchal, F., 2018. Optimal heat pump integration in industrial processes. Appl. Energy 219, 68–92.
Wallin, E., Berntsson, T., 1994. Integration of heat pumps in industrial processes. Heat Recovery Syst. CHP 14, 287–296.
Wolf, S., Flatau, R., Radgen, P., Blesl, M., 2017. Systematische Anwendung von Grosswärmepumpen in der Schweizer Industrie.
Zuberi, M.J.S., Bless, F., Chambers, J., Arpagaus, C., Bertsch, S.S., Patel, M.K., 2018. Excess heat recovery: an invisible energy resource for the Swiss industry sector. Appl. Energy 228, 390–408.